Tay noticed the characteristic red spot in the eye that is the earlier symptom of this disease due to degeneration in the central nervous system. Sach characterized this syndrome and identifying its familial characteristics, like mental disturbances, a deficiency in normal reflexes, progressive blindness and mortality, in Ashkenazi Jews (A Jewish ethnic group).

7.6.1 Causes

Tay-Sachs disease is classified as a lysosomal storage disease (Figure.7.9). In a healthy child, a lipid known as GM2 ganglioside, enter the nerve cell as a source of food. Among the components of the cell are lysosomal bags, which contain hexoaminidase A to digest this GM2 gangliosides. However, in Tay-Sachs disease, due to a failure in this degradation, levels of GM2-ganglioside increases in the lysosomes of neuronal tissue, thereby worsening the condition.

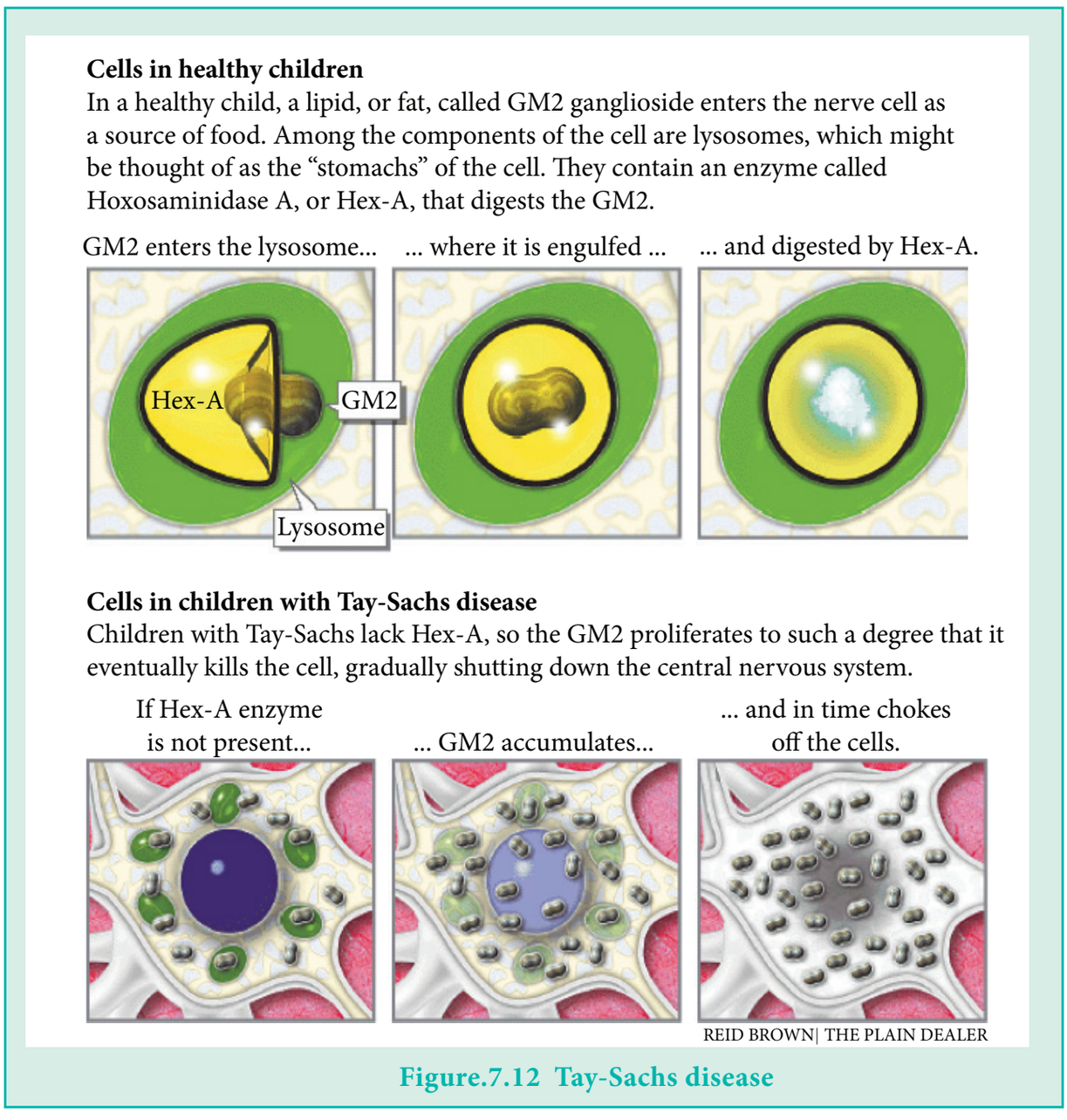


Figure.7.12 Tay-Sachs disease

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